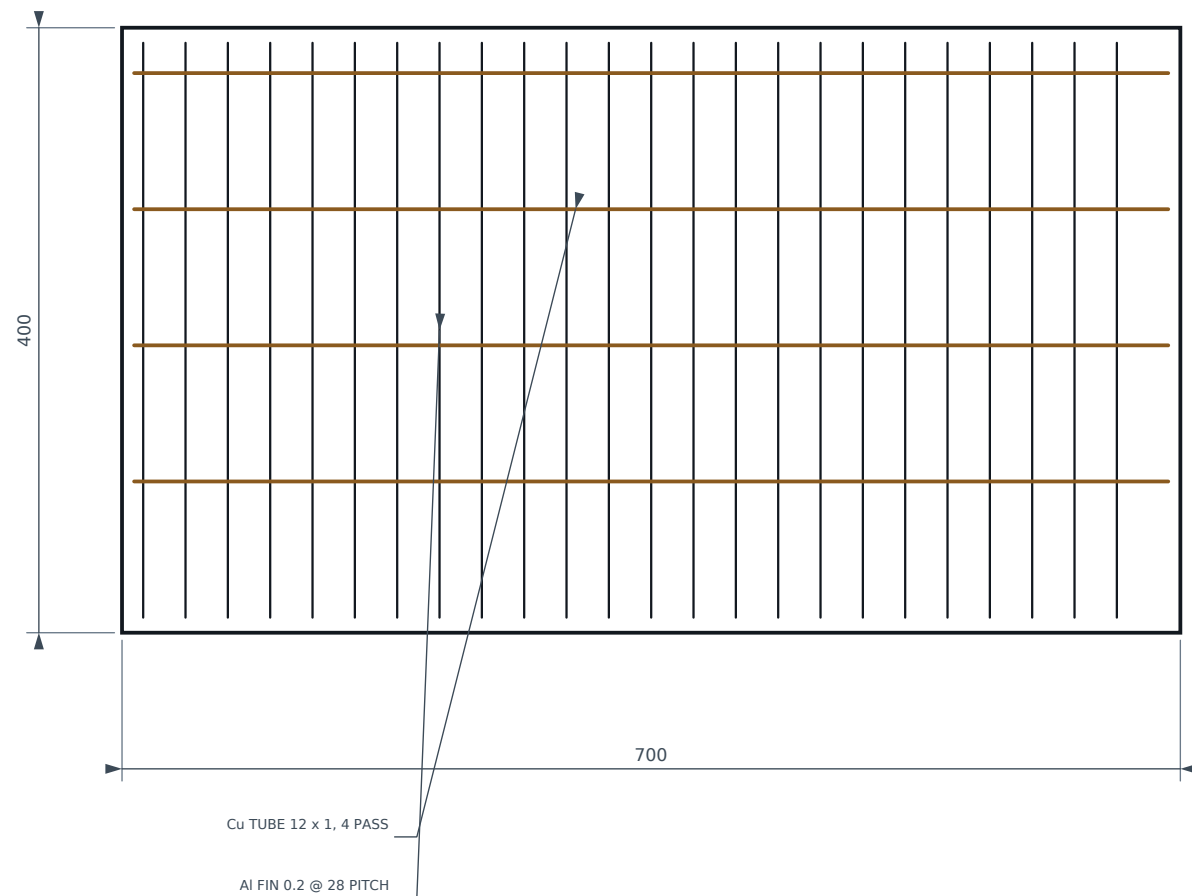


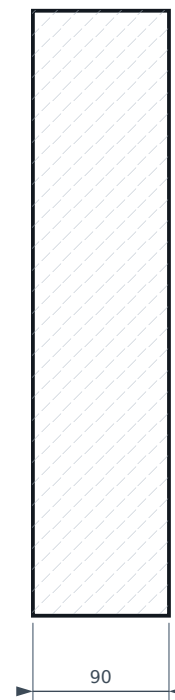
CONDENSER – FRONT

SCALE 1:5



SIDE

1:5

**RECEIVED SI-103**

Ø100

CONDENSER AND RECEIVER DATA			
condensing temperature			30 °C
condensing pressure			21.9 kPa abs
duty per day			3.76 MJ
mean power over 6 h			174 W
air-side dT			15 K
h, convection + radiation			10 W/m²K
surface required			1.16 m²
methanol condensed per day			3.19 kg
receiver working volume	0		3.0 L

NOTES

1. No fan. The condenser works on buoyancy and radiation, so it must be shaded from the sun and free to draw air from below.
2. Mount on the shaded side of the collector with 150 mm clear beneath and above.
3. Receiver sits below the condenser and above the evaporator: the loop is gravity-driven, there is no pump.
4. Cu tube to SS header joints brazed with Ag, then the whole assembly TIG welded into the envelope.
5. Fin pitch 28 is coarse on purpose. Natural convection chokes below about 6 mm free spacing.
6. Receiver has no sight glass and no drain. The charge is not serviceable in the field.

WORKING LEVEL 3.2 kg

INSPECTION		
CHECK	REQUIREMENT	METHOD
Helium leak, whole assembly	< 1e-8 Pa.m3/s	mass spectrometer
Proof pressure	21 bar, 5 min	hydrostatic, then dry
Condenser orientation	tubes horizontal, drain to receiver	level check
Shading	no direct sun on the fins	site survey

INDIA-VC THERMAL HARDWARE					ISO 128 / ISO 2768-mK	
TITLE CONDENSER AND RECEIVER AIR-COOLED, NATURAL CONVECTION / 174 W MEAN / 1.16 m2			SCALE 1:5 THIRD ANGLE PROJECTION		UNITS mm	
DRAWING No. SI-102 / SI-103			REV A		SHEET 3 / 6	
DRAWN AI-ASSISTED DRAFT	CHECKED — PENDING —	MATERIAL SEE BOM	FINISH SEE NOTES		DATE 2026-09-15	